

SOLID-PROPELLANT ROCKET ENGINES

Daire Joseph O'Sullivan

Cranfield University, Cranfield, Bedfordshire MK43 0AL, United Kingdom

Email: daire.osullivan@cranfield.ac.uk

Abstract

This paper has given a brief historical background into Solid Rocket Motors and established current demand for the technology. The basic operating theory behind SRM's has been explained along with key performance parameters. Mature market leaders have been identified. Promising startups have been surveyed and their relevance and strengths evaluated with reference to future trends. Solid Propellant Rocket motors are used throughout the defence and space industry due to their low-cost, long shelf life and simplicity. The influence of current geopolitics on these industries and their demand for SRM's has been discussed. SRM's are capable of delivering between 2 and 11 million Newtons of thrust which makes them suitable for a wide range of applications. SRM's therefore can be found in a diverse array of applications such as 1st stages for intercontinental ballistic missiles, deorbit motors on spacecraft and handheld missile launcher propulsion. The benefits and trade-offs of these many use-cases has been closely examined in this paper. In recent years new technologies such as additive manufacturing have been used to improve the design and manufacture of solid rocket motors. This paper has identified and evaluated these developments and attempted to predict future trends in Solid Rocket Motor technology and applications. There is also ongoing pressure for manufacturers to reduce their environmental impacts and this paper has examined the ongoing efforts.

Keywords: Space, propulsion, missiles, military.

Nomenclature

<i>ADN</i>	Ammonium Dinitramide
<i>AN</i>	Ammonium Nitrate
<i>AP</i>	Ammonium Perchlorate
<i>AM</i>	Additive Manufacturing
<i>DB</i>	Double-Base
<i>DLR</i>	German Aerospace Center
<i>HMX</i>	Cyclotetramethylene Tetranitramin
<i>HTPB</i>	Hydroxyl-terminated Polybutadiene
<i>ICMB</i>	Intercontinental Ballistic Missile
<i>I_{sp}</i>	Specific Impulse
<i>LRE</i>	Liquid Rocket Engine
<i>LRE</i>	Liquid Rocket Engine
<i>SLS</i>	Space Launch System

SRM

PLA

PPT

TVC

ULA

Solid Rocket Motor

Polylactic acid

Pulsed plasma thruster

Thrust Vector Control

United Launch Alliance

1. INTRODUCTION

Solid-propellant rocket engines commonly referred to as solid rocket motors (SRM) are rocket propulsion devices that utilise a solid fuel and oxygen mix to produce thrust. They are commonly used in the propulsion of munitions and space vehicles [1]. Their key advantages over liquid and hybrid rocket engines are their relative simplicity and reliability[2]. This is due to the lack of moving parts and the precast nature of the fuel. SRM's are the oldest type of rocket propulsion and can trace their origins back to ancient China with the invention of gunpowder[3]. This report is being carried out to assess and explain the nature of solid propellant engines, their history, working theory, use cases, advantages, disadvantages and future technological developments. SRM's are of particular interest in recent times due to the increasing warlike state of the world and the increasing interest in space exploration worldwide. Despite their lower performance when compared to liquid rocket engines, SRM's provide a low cost and reliable method to access space. Coupled with their unchallenged utility on the battlefield, this allows nations and companies to spread the cost of development across both space and defence budgets. This paper has reviewed several developments in SRM technology and their relevance today. The recent surge in demand for missile propulsion has meant that new solutions are being sought to reduce production times with additive manufacturing playing a key role. This demand has also allowed several new companies to enter the market. As with all technology in these times, efforts are underway to reduce the harm that the exhausts of SRM's cause with 'green' alternatives being investigated.

2. BACKGROUND AND KEY PLAYERS

2.1 History

The simplest form of SRM propellant is black powder which 15% charcoal, 10% sulphur and 75% potassium nitrate

[4]. Although initially used in harmless fireworks, their first reported use of a rocket in war was in 1232 by the Chinese against the Mongols[5]. Modern solid propellants as we know them, were first developed by Alfred Nobel in 1890 [4]. These would be classified as double-base propellants today. In 1942, researchers at Caltech experimented with asphalt and potassium perchlorate to create the first composite propellants [6]. Development of propellants has largely stagnated in the US since the 1970 however there has been continuous improvement in the materials and structures used in casings and nozzles[1][7].

2.2 Modern Context

In the modern day the weapons supply chain is facing huge challenges due to the consumption of munitions in the war in Ukraine. The Center for Strategic and International Studies has highlighted this high consumption rate and warned about the current time it would take to replenish supplies[8]. Concurrently, the space industry is currently experiencing a boom and demand for rocket launches are high[9]. United Launch Alliance's (ULA) Vulcan and Arianegroup's Ariane 6 are just two examples of launch vehicles that use large SRM's on their first stage[10]; [11]. Both rockets are aiming for over 10 launches a year to meet the market demand[12] [13]. Figure 1 shows an SRM being used to launch a payload, In the US, the SRM manufacturing base faces several challenges to meet this demand. In recent decades infrastructure has been allowed to age, environmental and safety regulation have grown, and a lack of development has led to brain drain[14].

2.3 Established Companies

The two key manufacturers of SRM's in the US are Aerojet-Rocketdyne division of L3Harris and Northrup Grumman[15]. In Europe Roxel, Nammo and Avio the lead the market. Although these manufacturers predominantly supply SRM's for use in defence applications, they are also the main providers for SRM's for use in space. Northrup Grumman produces the boosters as well as several other smaller SRM's that will be used to help send people back to the moon on NASA's Space Launch System (SLS)[16]. Avio has a 50% stake in EUROPROPULSION which manufactures the P120 motors used on the Vega and Ariane launch vehicles [17].

3. BASIC THEORY

3.1 Performance Characteristics

When discussing rocketry, the key performance characteristics are thrust, specific impulse(I_{sp}), effective exhaust velocity and mass fraction[1]. The thrust of a rocket is the force measured in Newtons created by the propulsion system[1]. The following definitions contextualise the utility of this thrust. The I_{sp} of a rocket is the thrust per a unit of flow rate with SI units of metres per second. It is common however to see the Imperial units being used which are seconds [19]. The I_{sp} is an indicator of how much propellant is needed to achieve a certain amount of thrust and can be compared to the miles

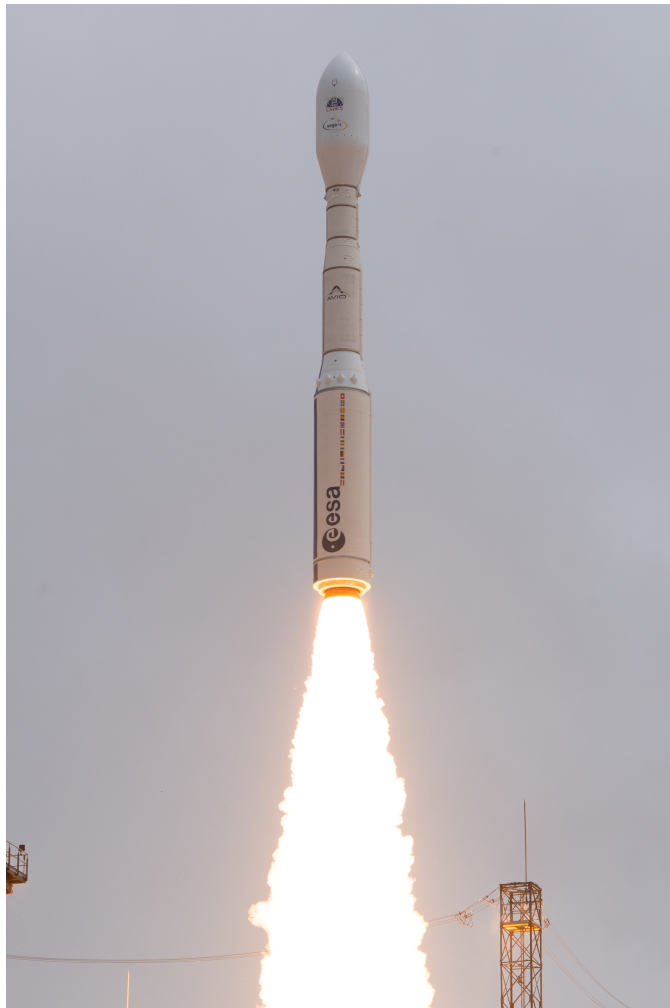


Fig. 1: The Vega-C which uses an SRM first stage lifting off from: [18]

per gallon of a car[1]; [2]). The effective exhaust velocity is a similar characteristic to I_{sp} and defines the average velocity of the rocket exhaust. Its units are also metres per second[1]. The mass fraction of the propulsion system is the percentage of the system made up of useable propellant. A high mass fraction suggests that the design is very structurally efficient with little inert mass[2].

3.2 Advantages and Disadvantages over LRE's

Liquid rocket engines using hydrogen and oxygen have the highest values of I_{sp} (up to 450 seconds) due to the low molecular weight of the propellants. However, hydrogens low density even at cryogenic temperatures requires large, insulated tanks which increase the mass fraction and complexity of the system(Edberg and Costa, 2022). In contrast SRM's have much lower values of I_{sp} , approximately 250 seconds. Mass fractions of over 90% can also be achieved by SRM's[19]. SRM's are also not limited by the amount of instantaneous thrust that they can produce as combustion takes place across the entire grain and is not limited by a fluid pump[19]. This has made SRM's an attractive choice for the first stages of

space launch vehicles. Many LRE launch vehicles such as the Atlas vehicles feature strap-on boosters to augment their thrust at launch[2]. SRM's can also be stored for 5-20 years before use and do not need to be fuelled before launch[1].

3.3 Key Components

An SRM usually consists of a metallic or composite pressure vessel into which the solid fuel known as a grain is cast. One or more mounting flanges can be attached to the casing to transfer thrust to the vehicle. This grain has a port in its centre where combustion takes place producing the hot gases. These gases are directed towards a nozzle which produces thrust. To initiate combustion an igniter is installed at the top of the grain. A layer of insulation is placed around the grain to prevent the pressure vessel overheating and losing structural integrity [1]. Figure 2 below illustrates the arrangement of these components.

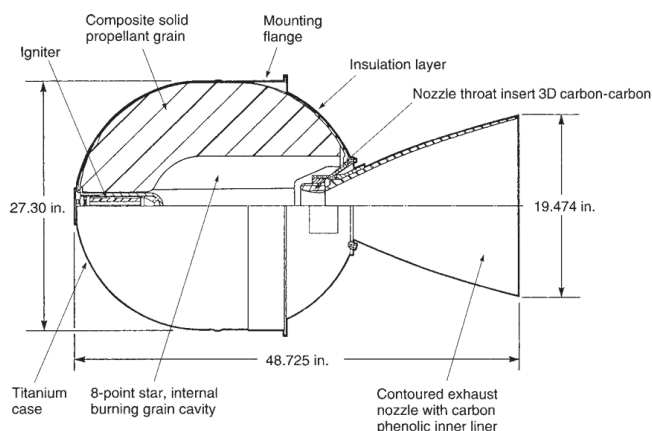


Fig. 2: SRM Cutaway from: [1]

3.3.1 Grain

Internal Ballistics Analysis is the term given to the prediction of chamber pressure in the motor as a function of time[19]. During combustion the burning face recedes perpendicular to itself. The rate at which the grain burns is proportional to the pressure within the combustion chamber. For a given propellant type the regression rate and thrust profile of an SRM is determined by the shape of port[1]. It is therefore possible to design the thrust profile according to mission requirements. SRM's can have progressive, neutral or regressive thrust profiles as shown in Figure 3 below. A neutral burn features a thrust that does not vary much over time. Regressive motors have thrusts that decrease over time and progressive motors have thrusts that increase. Neutral or regressive burns are usually desirable as the mass of the vehicle decreases as time goes on as does the atmospheric drag[19]. It is not uncommon to find unburnt remnants of propellants called "slivers" leftover after the burn due to the shape of the port[1].

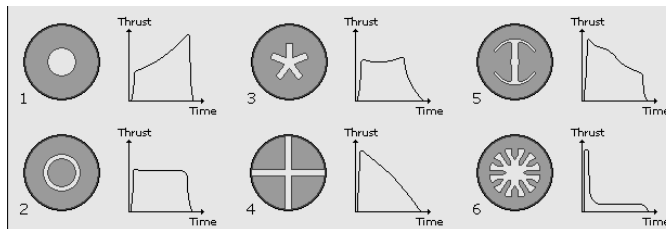


Fig. 3: Thrust Profiles from from: [20]

3.3.2 Nozzle

The primary function of the nozzle is to convert the hot gases created by combustion into useful thrust. Nozzles must be manufactured from materials that are capable of withstanding high temperatures. Due to the lack of a liquid propellant that can be used as a coolant, SRM nozzles must be cooled through ablation or radiation. Materials such as graphite, phenolic or 3D carbon-carbon are usually chosen for this purpose. Thrust vectoring control (TVC) can be achieved on SRM's at the nozzle (sutton) SRM's that feature TVC will usually achieve this at the nozzle. The addition of a TVC system can increase the mass of the nozzle by 50%. TVC can be achieved through the gimbaling of the nozzle such as on the Space Shuttle SRM or with liquid injection such as on the Titan IV boosters [19].

3.3.3 Insulation

The insulation in an SRM must protect the pressure vessel from the heat of combustion. The insulator is usually made up of a rubber with a fibre such as Kevlar pulp or asbestos. This rubber liner should char and create a protective layer that protects the virgin rubber beneath[1].

3.3.4 Igniter

The igniter of an SRM depends upon its size and can be as simple as an electric arc or another smaller SRM. Larger SRM's tend to use pyrogen ignition systems which while smaller SRM's will only use a pyrotechnic. An igniter can be thought of as a smaller SRM [2].

3.4 Propellant Composition

There are two main SRM fuel types, double base and composite. Double-base propellants are made of a homogenous mixture of nitrocellulose and nitro-glycerine. Both components contain a fuel and oxidiser and would ignite on their own. The nitrocellulose absorbs the nitro-glycerine which acts as a plasticizer. Double-base propellants often contain several other additives necessary for high performance. Carbon black is usually added to prevent excessive heat radiation through the propellant which could affect the burn rate [4]. Double-base propellants can be manufactured through solvent extrusion, solventless extrusion or casting. In the extrusion methods the propellant is forced through a die to form the desired shape. They differ in the addition of a solvent which can slightly dissolve the propellant[3]. Composite propellants are made up of an oxidiser, a fuel and a binder. The fuel is usually a powdered metal with the oxidiser consisting of a crystalline material and the binder a polymer or rubber[1].

Aluminium is commonly used as a fuel in combination with an Ammonium Perchlorate (AP) oxidiser and HTPB binder. Ammonium Nitrate (AN) is a lower performing oxidiser that can be used when less smoke is desired. PBAN is also used as a binder and was found on the Space Shuttle SRM's [19]. Composite propellants can be doped with catalysts or modifiers to increase the burn rate [1]. Copper and iron oxide are popular catalysts, but HMX can also be used. Catalysts are usually used to make up for a reduction of burn rate due to the addition of bonding agents and other chemicals that are essential for the structural integrity of the grain but detrimental to the burn rate [19]. Figure 4 below shows the relative performance of different propellant types. Although numerical models do exist to calculate the burn rate of SRM propellants, it is necessary to gather experimental data to accurately characterise a new propellant. These experiments can either be carried out in highly instrumented full-scale motors, subscale motors or in a "bomb" or Crawford burner [1]. Within a bomb, a small strand of the propellant is burnt within a pressurised inert gas [19].

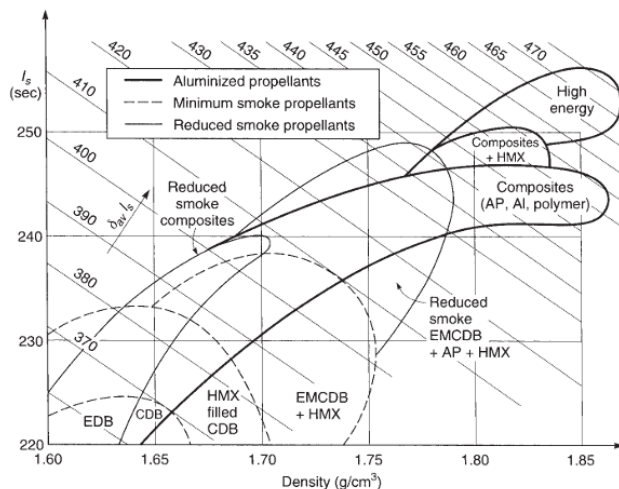


Fig. 4: Performance characteristics of different solid propellants from: [1]

3.5 Plumes

The plume is the exhaust of the SRM which contains the hot exhaust gases and unburnt elements of the fuel. AP based composite propellants generate large amounts of smoke and therefore AN composites are used in situations where visible plumes will be an issue. The use of double-base propellants is also common in these situations as well as the omission of aluminium. The downside of these methods is that the resulting propellant has lower performance. [1] Demand for smokeless propellants largely comes from military applications of SRM's. Afterburning of the exhaust gases once they have left the nozzle creates high infrared signatures and visible unburnt particles in the plume which make missiles highly detectable [21]. A long trail of smoke will direct any enemy directly to

the launch site. Large plumes can also extinguish the engine of the aircraft that the missile is fired from [21].

4. APPLICATIONS

4.1 Space and ICBM

ICBM's must be responsive, have short burn times and a low inert mass fraction [2]. SRM's have advantages in cost, safety and responsiveness over LRE's for ICBM propulsion [22]. The Minuteman I missile was 20% of the cost of the Titan I it replaced and required two instead of six crew to operate [22]. Legacy LRE ICBMs, such as the Titan missiles, tended to use highly toxic hypergolic storable propellants. These propellants were difficult to store safely for long periods of time and there have been catastrophic accidents as a result [23]. When the Titan II was deactivated in 1987, the US ICBM fleet became completely dependent on SRM for propulsion [22]. SRM's have similar advantages for space launch vehicles as they do for ICBMs, however they have not become the default choice. Low values of I_{sp} , the inability to test before launch and a lack of throttling are detrimental to SRM's utility, especially when carrying humans [24]. Historically, SRMs have also been used in apogee boosting and orbit changing of spacecraft and scientific probes [19]. Notable examples of this are the Magellan and Galileo probes, the former of which used two SRM's, one to launch from the space shuttle and a second to enter into orbit around Venus [25]. In this use cases, compatibility and high density win out over the performance of LRE's [19]. Due to the lack of vectoring in these systems, they are often spin stabilized. This spin stabilisation often needs to be supplemented by additional manoeuvres and the spacecraft must be designed to withstand the high rotational velocity [26]. The similarities between ICBM requirements and space launch are such that decommissioned ICBM's have often formed the basis of new launch vehicles and upper stages such as with the Minuteman missile [27]. One of the strongest factors driving SRM usage in space applications is the requirement to retain knowledge and keep production lines open for SRM's in case of war.

4.2 Missile Systems

SRMs are used in a range of missile systems from shoulder launched to the ICBM's discussed above. SRMs are particularly desirable in weapons systems due to their ruggedness, storability and instantaneous thrust. SRM's used in missiles often feature a blast tube to enable the fitting of fins and their actuators at the base of the missile. The blast tube also serves to stabilise the centre of gravity of the missile [1].

4.3 Sounding Rockets and Hobby Rocketry

The recently developed Red Kite motor for the DLR points towards another driver of SRM development. Sounding rockets are used to lift scientific payloads on suborbital parabolic trajectories [28]. Sounding rockets usually take advantage of military supply chains to provide SRM's. The Red Kite motor was developed specifically to be a commercial option free

from any restrictions[29]. There also exists a market for the use of SRM's in student and enthusiast rocketry. The main manufacturers in this area are Aerotech and Cessaroni[30]. Even here the impact of war can be seen as modifications of these motors for use in military applications has been examined [31].

5. FUTURE DEVELOPMENTS TRENDS AND NEW MARKET ENTRANTS

5.1 New Companies

With the recent shortfall in SRM manufacturing capability, several new companies have entered the market. Notable among these are Anduril, X-Bow Systems and Ursa Major[15] [32]. These companies are leveraging new technologies to give them a competitive advantage over the incumbents.

5.2 Additive Manufacturing

It is important to distinguish between the use of AM in vessel and grain manufacturing. The use of additive manufacturing for metal and plastic component is well proven and can be found in many off-the shelf products today. Ursa Major are leveraging this technology to quickly manufacture their new line of Lynx SRM's [33]. They claim that by using this technology they can rapidly scale up their manufacturing lines to meet the current demand. It will also allow them to iteratively improve the design of their motors.

There is much ongoing research into the use of additive manufacturing (AM) to produce the actual fuel grains in SRM's. This technology could allow for novel port shapes, embedded devices and complex propellant layering [34]. These additions would allow for a new range of thrust profiles and missions. Embedded strain gauges can be used to monitor the condition of the motors during storage which will improve longevity and safety. Altering propellant composition layer by layer can allow the SRM to vary its thrust in accordance with the stage of flight with more precision than casting currently allows. The same can be said for the port shape. Experiments have already shown that photocurable binders suitable for AM can have better mechanical properties than HTBP[34]. PLA-AP propellant mixes have also been successfully printed[35]. This technology has the potential to reduce the barrier to access for new SRM manufacturers. One of the main challenges that AM for SRM's must overcome is developing new printers capable of extruding the dense slurries. Vibration assisted printing is a technique that uses ultrasound to temporarily reduce the density of the slurry and could be an enabling technology[36].

X-Bow is one company that is aiming to leverage this emerging technology to rapidly manufacture SRM's to meet the current demand and has already secured a contract with the US Department of Defense[37], [38] .

5.3 Electric and Extinguishable Solid Propulsion

One particularly interesting area of research with far reaching potential is that of electrically ignited solid motors

or extinguishable solid propellants. These motors are made up of an AP propellant with additives that reduce ignition temperatures. This allows the motor to be ignited and throttled using an electrical current [39]. When the current is removed the motor is extinguished. These motors are also immune to spark ignition, bullets and impacts leading to greater safety. These motors can be used as igniters for larger motors, providing a safer alternative to conventional pyrotechnics[40]. These motors have applications in both defence and space. They differ from PPT systems in that they produce thrust through combustion not the production of a plasma, however they can be used in a dual mode if required [39].

Raytheon has explored the use of this technology in both space and defence applications. They propose that it can provide high delta V propulsion for small satellites, a market which is currently unoccupied [41]. The technology can be utilised to provide better control of the thrust throughout the flight as segments of the motor can be burned individually.

5.4 Green Propellants

The combustion gases of AP based propellants contain high amounts of hydrochloric acid which when released in the upper atmosphere can lead to acid rain and depletion of the ozone layer[42]. It has also been linked to thyroid gland interference[25]. ESA's Clean Space Initiative and the Green Propulsion Harmonisation Process has also put pressure on companies to find "greener" alternatives to AP[43]. For tactical uses the smoky nature of AP's exhaust is also undesirable. Ammonium Nitrate and Ammonium Dinitramide are two of the alternatives being examined[43]. Ammonium Nitrate is cheap and commonly available as it is used as a fertilizer and produces little smoke as it burns[42]. However, it is hygroscopic and does not have the performance of AP. It's use in the past has been confined to gas generators[43]. ADN has high performance however it is highly sensitive and therefore would not make a good replacement for AP. A promising solution is to use a mix of ADN and AN to create a green, high performing, low sensitivity propellant . Research into this propellant has shown that values of I_{sp} similar to that of AP propellants are possible.

6. CONCLUSION

SRM's are a robust and well-developed method of generating thrust for both space and tactical usecases. Although not as versatile as LRE's due to the lack of throttling, their simplicity has meant that demand remains for the technology.

The predominant justification for development and manufacture SRM's come from the defence industry. LRE's are incompatible with the need for responsiveness and survivability. The utilisation of SRM's in space is a by product of the technology development and manufacturing line developed for military use.

In the US new startups are emerging to meet the demand for SRM's, leveraging new technologies to gain an edge over incumbent manufacturers. War in Ukraine has bolstered the industry and woken it up from decades of stagnation

An emphasis has been placed on American history and development due to the availability of resources, however it can be assumed that the challenges are common across the industry. Further research into the Russian and European SRM industries would be desirable.

There is a clear new interest in solid rocket motors and much research is being completed to overcome their disadvantages. These include green propellants and throttling or the ability to extinguish them.

REFERENCES

- [1] George P. Sutton and Oscar Biblarz. *Rocket Propulsion Elements*. Ninth. John Wiley Sons, Inc, 2017. ISBN: 9781118753651.
- [2] Stephen D. Heister et al. *Rocket Propulsion*. 1st. Cambridge University Press, 2019.
- [3] Clayton Huggets, C.E. Bartley, and Mark M. Mills. *Solid Propellant Rockets*. 1st. Princeton University Press, 1960.
- [4] Morton Shorr and Alfred J. Zaehring. *Solid Rocket Technology*. 1st. John Wiley & Sons, Inc, 1967.
- [5] Nancy Hall. *Solid Rocket Engine*. May 2021. URL: <https://www.grc.nasa.gov/www/k-12/airplane/srockth.html>.
- [6] J. D. Hunley. "The history of solid-propellant rocketry: What we do and do not know". In: *35th Joint Propulsion Conference and Exhibit*. American Institute of Aeronautics and Astronautics Inc, AIAA, June 1999. DOI: 10.2514/6.1999-2925.
- [7] Mark J. Lewis and John Fischer. *Emerging Technology Horizons: U.S. Needs to Refocus on Energetic Materials*. Accessed on 03, 12, 2025. URL: <https://apple.news/AFIoSOAZWQEgmXqkY8HqQrQ>.
- [8] Mark F. Cancian. *Rebuilding U.S. Inventories: Six Critical Systems*. Jan. 2023. URL: <https://www.csis.org/analysis/rebuilding-us-inventories-six-critical-systems>.
- [9] Jacqueline Feldscher. *The State of Launch*. Accessed on 03, 12, 2025. Apr. 2024. URL: <https://payloadspace.com/the-state-of-launch/>.
- [10] ULA. *Vulcan*. Accessed on 03, 12, 2025. URL: <https://www.ulalaunch.com/rockets/vulcan-centaur>.
- [11] ESA. *Ariane 6 overview*. Accessed on 03, 12, 2025. URL: https://www.esa.int/Enabling_Support/Space_Transportation/Launch_vehicles/Ariane_6_overview.
- [12] Joey Roulette. *Vulcan rocket's debut brings long-awaited challenge to SpaceX dominance*. Accessed on 03, 12, 2025. Jan. 2024. URL: <https://www.reuters.com/technology/space/vulcan-rockets-debut-brings-long-awaited-challenge-spacex-dominance-2024-01-10/>.
- [13] Jeff Foust. *Arianespace sees stronger institutional demand for Ariane 6 amid shifting geopolitics*. Accessed on 03, 12, 2025. Mar. 2025. URL: <https://spacenews.com/arianespace-sees-stronger-institutional-demand-for-ariane-6-amid-shifting-geopolitics/>.
- [14] Nadia Schadow et al. *Rocket's Red Glare: Modernizing America's Energetics Enterprise*. Tech. rep. Hudson Institute, Oct. 2022. URL: www.hudson.org.
- [15] Stephen Losey. *Businesses reposition amid growing demand for solid rocket motors*. Oct. 2023. URL: <https://www.defensenews.com/industry/2023/10/31/businesses-reposition-amid-growing-demand-for-solid-rocket-motors/>.
- [16] NG. *Artemis*. Accessed on 03, 12, 2025. URL: <https://www.northropgrumman.com/space/nasas-artemis-program>.
- [17] avio. *Successful First Test Firing for the P120C Solid Rocket Motor for Ariane 6 and Vega-C*. Accessed on 03, 12, 2025. July 2018. URL: <https://www.avio.com/index.php/press-release/successful-test-firing-p120c-solid-rocket-motor>.
- [18] esa. *Vega-c liftoff*. Accessed on 03, 13, 2025. 2022. URL: https://www.esa.int/Enabling_Support/Space_Transportation/Vega/Vega-C.
- [19] Ronald W. Humble, Gary N. Henry, and Wiley J. Larson, eds. *Space Propulsion Analysis and Design*. 1st. The McGraw-Hill Companies, Inc, 1995.
- [20] Brauenig. *Rocket and Space technology*. Accessed on 03, 12, 2025. 2012. URL: <http://brauenig.us/space/index.htm>.
- [21] Paul Zarchan, Gordon E. Jensen, and David W. Netzer, eds. *Tactical Missile Propulsion*. Vol. 170. American Institute of Aeronautics and Astronautics, 1996.
- [22] Tony C. Lin. "Development of U.S. Air Force Intercontinental Ballistic Missile Weapon Systems". In: *Journal of Spacecraft and Rockets* 40 (4 July 2003), pp. 491–509. ISSN: 0022-4650. DOI: 10.2514/2.3990.
- [23] Mark K. Christ. *Titan II Missile Explosion (1980)*. Accessed on the 05/03/2025 at encyclopediaofarkansas.net/entries/titan-ii-missile-explosion-2543/. Feb. 2025. URL: encyclopediaofarkansas.net/entries/titan-ii-missile-explosion-2543/.
- [24] Don Edberg and Willie Costa. *Design of Rockets and Space Launch Vehicles*. 2nd. American Institute of Aeronautics and Astronautics, Inc, 2022.
- [25] Niklas Wingborg. *Final Report Summary - HISP (High Performance Solid Propellants for In-Space Propulsion)*. Jan. 2015. URL: <https://cordis.europa.eu/project/id/262099/reporting>.
- [26] Charles D. Brown. *Elements of Spacecraft Design*. 1st ed. American Institute of Aeronautics and Astronautics, Inc, 2002.
- [27] E. E. Day. "Qualification of Minuteman Stage III Motor for Spin Stabilized Perigree Stages". In: *AIAA Paper*. AIAA, 1980. DOI: 10.2514/6.1980-1269.
- [28] NASA. *About Sounding Rockets*. Accessed on 03, 12, 2025. URL: <https://www.nasa.gov/soundingrockets/overview/>.
- [29] F Scheuerpflug et al. *The Red Kite Sounding Rocket Motor-Qualification Milestones and Application Spec-*

- trum The Red Kite Sounding Rocket Motor Qualification Milestones and Application Spectrum*. Tech. rep. DLR, 2024.
- [30] Apogee Components. *Rocket Motor Basics*. Accessed on 03, 12, 2025. URL: <https://www.apogeerockets.com/Rocket-Motor-Basics-Quick-Start-Guide?m=quickside>.
- [31] Nathan D. Stuffle. "Modification of Commercial Rocket Motors for Tactical Applications". PhD thesis. Naval Postgraduate School, 2022.
- [32] Audrey Decker. *New solid-rocket maker aims to start production next year*. Accessed on 03, 12, 2025. Nov. 2024. URL: <https://www.defenseone.com/business/2024/11/new-solid-rocket-maker-aims-start-production-next-year/400758/>.
- [33] *Ursa Major's Lynx: Pioneering the Future of Solid Rocket Motor Manufacturing*. Nov. 2023. URL: <https://www.ursamajor.com/blog/ursa-majors-lynx-pioneering-the-future-of-solid-rocket-motor-manufacturing>.
- [34] Justin Lajoie, Jacob Blocker, and Travis Sippel. "Rheological, Ballistic, and Mechanical Properties of 3D Printed, Photocured Composite Propellants". In: *Journal of Propulsion and Power* 39 (6 Nov. 2023), pp. 936–944. ISSN: 15333876. DOI: 10.2514/1.B39113.
- [35] Konstantin A. Monogarov, Igor V. Fomenkov, and Alla N. Pivkina. "FDM 3D printing of combustible structures: First results". In: *Mendelev Communications* 32 (2 Mar. 2022), pp. 228–230. ISSN: 09599436. DOI: 10.1016/j.mencom.2022.03.025.
- [36] Monique S. McClain et al. "Investigation of additively manufactured layered composite solid propellant". In: *AIAA Scitech 2020 Forum*. Vol. 1 PartF. American Institute of Aeronautics and Astronautics Inc, AIAA, 2020. ISBN: 9781624105951. DOI: 10.2514/6.2020-1427.
- [37] X-Bow. *We Make*. Accessed on 03, 13, 2025. URL: <https://www.xbowsystems.com/we-make/>.
- [38] Audrey Decker. *How X-Bow plans to break the Northrop-Aerojet 'duopoly' on rocket motors*. Accessed on 03, 12, 2025. Oct. 2023. URL: <https://www.defenseone.com/business/2023/10/how-x-bow-plans-break-northrop-aerojet-duopoly-rocket-motor-production/391492/>.
- [39] Matthew S Glascock, Joshua L Rovey, and Kurt A Polzin. *Specific Impulse of Electric Solid Propellant in an Electrothermal Ablation-fed Pulsed Plasma Thruster*. Tech. rep. University of Vienna, 2019.
- [40] W. Sawka and M. McPherson. "Electrical solid propellants: A safe, micro to macro propulsion technology". In: *49th AIAA/ASME/SAE/ASEE Joint Propulsion Conference*. Vol. 1 PartF. American Institute of Aeronautics and Astronautics Inc., 2013. ISBN: 9781624102226. DOI: 10.2514/6.2013-4168.
- [41] Frederick B. Koehler, Mark J. Meisner, and Jeff L. Vollin. "Multi-pulse solid rocket motor technology". In: *AIAA Propulsion and Energy 2020 Forum*. American Institute of Aeronautics and Astronautics Inc, AIAA, 2020, pp. 1–6. ISBN: 9781624106026. DOI: 10.2514/6.2020-3930.
- [42] Christopher Glaser, Pietro Bosco, and Andreas Ohndorf. "Green Solid Propellants: Trends, Perspectives and Challenges". In: *AIAA SCITECH 2025 Forum*. American Institute of Aeronautics and Astronautics, Jan. 2025. ISBN: 978-1-62410-723-8. DOI: 10.2514/6.2025-2686. URL: <https://arc.aiaa.org/doi/10.2514/6.2025-2686>.
- [43] N Wingborg et al. "GRAIL: Green Solid Propellants for Launchers". In: *EUCASS (2017)*. DOI: 10.13009/EUCASS2017-560.